A Fail-Closed Certificate Architecture for Literal Gaussian-Prime-Ideal Ownership in a Level-(3) Bianchi Flow

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Abstract

This article formulates a fail-closed certificate architecture for asking whether primitive loxodromic owners in a fixed level-(3) Gaussian Bianchi geodesic flow can receive one literal Gaussian prime ideal. It is a review-adjudicated synthesis of a frozen 22-source corpus, rather than a theorem of ownership, a mechanism implementation, a quotient computation, or a finite-refinement experiment. The corpus identifies the Picard/Bianchi object, primitive and inversion-aware terminology, adjacent arithmetic-group algorithms, and exact ideal-arithmetic components. It does not provide either the project-specific owner law or the complete primitive-root, conjugacy, inversion, and canonical-unoriented quotient required for evaluation. The literal one-ideal codomain is therefore treated as a deliberately strict stress test, not as a canonical or literature-forced output. We separate two open interfaces: Gate M for mechanism admissibility and Gate Q for quotient completeness. Their independence means only that neither gate entails the other at the design level. Downstream collision performance is firewalled from selecting or repairing either interface. The resulting prospective protocol specifies typed serializers, positive and negative certificates, immutable mechanism registration, controlled stop states, and independent replay. Because every inherited citation lacks a passage locator, claim-to-passage support remains inconclusive. No owner, quotient, statistic, or route advancement is reported.

Keywords: Bianchi geodesic flow; Gaussian prime ideals; primitive conjugacy classes; certificate architecture; strict codomain; evidence synthesis

Traditional Chinese Abstract / 繁體中文摘要

本文提出一套失敗即封閉的證書架構，用以檢驗固定層級三之高斯 Bianchi 測地流中，原始雙曲所有者是否能被一致地指派一個字面上的高斯素理想。研究僅綜整已凍結的二十二項來源及既有評審紀錄，不執行新定理證明、機制建置、商集計算或有限精化實驗。文獻足以界定 Picard／Bianchi 幾何對象、原始與反向辨識語彙、相鄰算術群演算法及理想算術工具，卻未提供本計畫專屬的所有者法則，也未完成最大根、共軛、取逆及無向正規化所需的完整商集。故單一字面素理想被明確視為刻意嚴格的壓力測試，而非典範、伽羅瓦穩定或由文獻必然導出的值域。本文把機制可容許性與商集完備性分成兩道仍開放的關卡；兩者僅在設計蘊涵上互不替代，下游碰撞分離表現不得用來選擇或修補任一關卡。架構進一步規定型別化序列化、正反證書、不可變註冊、停止狀態與獨立重播。所有引文均缺段落定位，故支持狀態仍為不確定；本文不宣稱任何所有者、商集、統計值或路線晉級。

關鍵詞： Bianchi 測地流；高斯素理想；原始共軛類；證書架構；嚴格值域；證據綜整

1 Introduction, Research Question, and Contribution Boundary

The study fixes one mathematical frame before performance is visible. Its dynamical object is the inherited unit-speed geodesic flow on a torsion-free level-(3) Gaussian Bianchi manifold. Hyperbolic arclength is the clock. A prospective owner is a primitive loxodromic conjugacy class; inversion identifies the primary unoriented owner, and powers are repetitions rather than new owners. The requested arithmetic output is one literal nonzero prime ideal of `Z[i]`. Associates denote the same ideal, while conjugate prime ideals above a split rational prime remain distinct. These are frozen design definitions, not findings extracted from the corpus.

The research question is deliberately conditional: under that strict output type, what must be certified before a performance-independently registered owner mechanism could be evaluated on a certified finite owner population? The question does not assume that a lawful mechanism exists. It also does not ask whether a norm, rational prime, composite ideal, residue class, or unordered conjugate pair might be a useful alternative object. Those alternatives are relevant to the interpretation of the frame, but substituting one of them after review would answer a different question.

The central distinction is between two pre-performance obligations. The mechanism-admissibility obligation asks whether a frozen formula maps each admissible primitive unoriented owner to one literal Gaussian prime ideal while respecting conjugacy, inversion, and repetition. The quotient-completeness obligation asks whether the inherited finite matrix population has been converted into exact primitive roots and canonical unoriented classes with complete positive and negative certificates. They are design-level independent: a formula cannot decide the quotient merely by producing labels, and a complete quotient cannot invent or validate a formula. This is not a proposition that the two failure events are statistically independent, nor a theorem that every possible construction separates into these modules.

The article's contribution is accordingly procedural and falsifiable. It specifies a fail-closed sequence in which neither classification performance nor collision separation can select a mechanism, repair an invalid codomain map, or excuse an incomplete owner quotient. A mechanism can fail; a quotient can remain unresolved; the downstream estimand can remain not evaluable. These are valid outcomes. The architecture is not claimed to be novel, and it is not presented as an existence or nonexistence theorem about intrinsic arithmetic ownership.

1.1 Claim classes and failure semantics

Four kinds of statements must remain distinct throughout the article. First, the frozen object definitions specify what this study elects to mean by flow, clock, primitive owner, repetition, inversion, and arithmetic output. They make the question determinate, but they do not show that an owner mechanism exists. Second, corpus findings describe only what the bounded evidence package can support at its present resolution. A source may identify the relevant group, discuss primitive geodesics, or provide an adjacent algorithm without instantiating the map required here. Third, architectural requirements state what a future implementation would have to serialize and verify. They are normative proof obligations, not completed constructions. Fourth, status statements record what has actually happened: no mechanism has been admitted, no quotient has been certified, and no downstream statistic is available.

This classification prevents several common but consequential substitutions. A well-defined ideal factorization algorithm does not select an ideal for a geometric owner. A conjugacy routine for a neighboring input class is not automatically a solver for the frozen level-(3) relation. An invariant observed on matrix rows does not turn those rows into primitive unoriented owners. Conversely, failure to find a mechanism in a bounded search would not prove nonexistence. Each statement must be evaluated against its own input type, output type, witness, and quantifier.

The literal single-ideal codomain is kept fixed precisely because it is a stress test. The paper asks whether an unoriented primitive owner contains enough representative-independent information to select one literal Gaussian prime ideal, with associates identified but conjugate split primes still distinct. A later choice to return a norm, rational prime, composite ideal, residue class, or unordered conjugate pair may define an interesting study, but it would alter the frozen output type. Such a change cannot be treated as a repair after the present mechanism or performance has been inspected. Keeping the strict target does not elevate it to a canonical one; it preserves the falsifiability of this particular question.

Fail-closed semantics give informative meaning to an incomplete pipeline. Missing data do not default to a negative theorem, and an unresolved relation does not default to inequality. Gate M can remain unevaluable because no formula has been frozen, or it can refute a particular frozen formula by a reproducible law violation. Gate Q can remain unevaluable because its exact decision procedures are absent, or stop unresolved under a prospectively fixed resource limit. These outcomes are more precise than a generic failure label. They identify which interface lacks evidence without granting conclusions about other mechanisms, broader codomains, or the global arithmetic of closed geodesics.

2 Frozen Literature and Theoretical Frame

2.1 The Picard/Bianchi object does not itself supply an owner law

Prime-geodesic work on the Picard manifold supplies direct context for the quotient associated with `PSL_2(Z[i])`, but the frozen verification record admits it only as object-level and analytic context, not as a class-to-ideal construction [3].

A second study treats the prime geodesic theorem for the same Gaussian group and spectral exponential sums without thereby defining the project's owner map [14].

Gaussian arithmetic and ideals occur naturally in analysis of the Picard setting, yet the verified support boundary does not turn that shared arithmetic vocabulary into a unique literal-ideal assignment [4].

Primitive closed-geodesic and conjugacy-class language, including inversion-aware or unoriented formulations in a nearby arithmetic hyperbolic setting, helps state the owner type while leaving the exact noncompact level-(3) decision procedure open [8].

Earlier Picard prime-geodesic work adds historical analytic context but no owner law [15].

Hyperbolic-three-manifold explicit-formula work similarly broadens the analytic setting without supplying the required project map [18].

The correction-bound analytic record is retained with a strict provenance distinction. The original hyperbolic-three-manifold article cannot be used for an affected statement without its correction and addendum [1].

The paired correction is part of the evidence boundary, not an additional mechanism source and not a retraction-clearance record [2].

A current first-party preprint concerns the exact Gaussian quotient but remains a preprint and does not close the mechanism or quotient interfaces [21].

Arithmetic-hyperbolic background gives definitions and comparison context for Kleinian groups, trace fields, and hyperbolic three-manifolds, but the frozen source assessment admits no direct owner formula from it [17].

Bianchi and Picard actions and imaginary-quadratic arithmetic are likewise part of the structural setting rather than evidence that a literal ideal owner is canonical [10].

Taken together, these records support the registered distinction between knowing the object and possessing the project-specific owner law.

2.2 Quotient completeness is a composite certificate target

The frozen algorithmic sources offer components with different input and output types. Arithmetic-Kleinian fundamental-domain computation can inform exact object representations and reduction witnesses, but the source-verification boundary does not treat it as an implemented level-(3) primitive-unoriented quotient [19].

General algorithms for arithmetic groups motivate decidability and certificate questions while leaving the exact project relation and serialization to be established [12].

Primary conjugacy methodology for certain arithmetic groups is an adjacent component, not evidence that the frozen subgroup solver has been instantiated [13].

Presentations for special linear and Bianchi cases can support a word-to-matrix interface but do not alone provide maximal-root, conjugacy, inversion, and canonicalization certificates [22].

Nearby conjugacy literatures sharpen the no-transfer rule. An algorithm for groups of oriented geometrizable three-manifolds has differently typed hypotheses and cannot be imported without an exact reduction to the frozen group [20].

A word-hyperbolic conjugacy result is similarly conditional on a hypothesis class that is not itself the project certificate [11].

Constructive integral-matrix conjugacy in `GL(n,Z)` can inform witness design, but equality in that ambient matrix group is not automatically the required conjugacy relation in the level-(3) projective subgroup [9].

The quotient target therefore remains a composition problem. It requires an exact projective-lift convention, loxodromic eligibility, maximal group-element roots, level-(3) conjugacy, inversion links, deterministic canonical unoriented identifiers, transitivity checks, and population accounting. A bounded search that does not find a conjugator is not a negative certificate. The corpus supplies possible ingredients but no one frozen source supplies this complete target.

2.3 Ideal arithmetic validates outputs but does not choose owners

Algorithms for number fields and ideal factorization support later validation of whether a serialized output is a nonzero Gaussian prime ideal and how rational primes split in `Z[i]`; they do not identify which class should own which ideal [7].

Effective algebraic-number-theory methods support exact arithmetic and certificate-oriented implementation, again without providing a class-to-ideal selection law [16].

Practical computational number-theory interfaces clarify what an implementation would need to serialize and verify, but not why a particular split branch belongs to an unoriented owner [5].

Maximal-power detection for integral ideals provides an ideal witness and must not be conflated with maximal roots of group elements or with geodesic repetition [6].

The literal codomain is thus a deliberately strict frozen frame. It is not represented as Galois-stable, canonical, or forced by the literature. Because an unoriented owner suppresses an orientation-like choice while a split rational prime yields two distinct conjugate prime ideals, a candidate mechanism must justify a representative-independent literal branch under the frozen question. Failure would be a conditional strict-frame obstruction. It would diagnose incompatibility between that candidate, the owner semantics, and the selected output type; it would not show that all broader notions of arithmetic ownership are impossible.

3 Executed Methodology: Frozen Evidence Synthesis Only

The executed research method was the already completed, bounded literature workflow underlying the present article. Phase 2 inspected 48 record manifestations, removed 12 duplicate manifestations, screened 36 unique records at title or abstract level, and admitted 22 unique records after scope screening. Seventeen admitted records were counted within the workflow's peer-reviewed journal-or-correction numerator. The source inventory contains 22 unique identifiers, P29-S01 through P29-S22. Independent verification recorded 22 `VERIFIED` existence outcomes and bounded metadata normalization. These counts describe the frozen workflow; the present composition did not repeat or extend the search.

Phase 3 encoded all 22 records in a source-effect matrix with support class, claim-fitness grade, admissible use, prohibited stronger use, compatibility role, and locator limitation. Phase 4 compiled that matrix into a research report. Phase 5 then supplied four procedurally separated reviews and a synthesis checkpoint. The author-adjudicated Phase-6 materials narrow and clarify the registered claim surfaces used here. No new database, website, publisher page, full text, passage, quotation, scientific artifact, or experiment was retrieved or created.

The method has an explicit evidentiary ceiling. In this paper, all 22 inherited prose citation pairs retain `anchor:none`; across the five-paper Round-10 batch, all 144 inherited citation pairs likewise have no passage locator. Exact theorem pages, sections, paragraph identifiers, and quotation spans were not frozen. Source identity, DOI/reference closure, metadata verification, and bounded abstract or scope inspection do not establish claim-to-passage faithfulness. The claim-to-passage status for every cited premise in this article is therefore `INCONCLUSIVE`. No direct quotation appears.

Correction, source-status, and integrity checks remain separately typed. P29-S06 stays bound to P29-S07 for affected use. P29-S09 remains a preprint. Structured retraction status is `NOT_CHECKED` for all 22 source rows, and source-level conflict-of-interest status is `UNKNOWN_NOT_AUDITED`. The absence of a fabricated or unverifiable inventory row is not retraction clearance, and a correction binding is not a full theorem-applicability audit.

3.1 Transformation from evidence records to manuscript claims

Stage 2 converts the frozen evidence synthesis into a publication-shaped argument without increasing its scientific strength. Each registered claim has a preassigned source set and an explicit negative constraint. The manuscript may narrow a claim, explain its logic, or state a prospective certificate field, but it may not add an unassigned source, infer a stronger theorem, or turn an intended interface into an executed result. This rule is especially important where several adjacent sources appear to compose naturally: composition at the level of a research plan is not composition at the level of a proved terminating algorithm.

Source identity is also kept separate from passage support. The source inventory, DOI closure, bounded metadata checks, and support-role coding justify citing a record for a narrow contextual purpose. They do not identify an exact theorem statement or discharge its hypotheses. For that reason every citation is paired with the same unresolved passage status. The correction relation between P29-S06 and P29-S07 is retained, and the preprint status of P29-S09 remains visible, but neither condition is converted into a general integrity clearance. This is a deliberate limit of the present manuscript rather than a silent assumption.

The executed outcome of this transformation is therefore modest but concrete: a coherent argument, a closed bibliography with all 22 frozen identifiers, and an auditable set of claims whose prohibited upgrades remain legible. No numerical or algebraic computation is hidden inside the prose conversion. No candidate formula, matrix population, quotient ledger, collision table, or replay receipt was generated. The article can guide a later scientific stage, but it cannot serve as evidence that the later stage has already occurred.

4 Review-Adjudicated Proof and Method Architecture

4.1 Gate M: mechanism admissibility under the strict frame

Gate M is a prospective logical and serialization contract. Before any collision-separation outcome is visible, a candidate mechanism would have to be registered by immutable formula bytes, source trace, admissible domain, exact codomain, normalization, treatment of associates, and deterministic behavior under conjugacy, inversion, and repetition. Its output validator would have to distinguish a literal Gaussian prime ideal from a norm, a rational prime, a composite ideal, a residue, or an unordered conjugate pair. A split branch must follow from the registered mechanism rather than from the observed finite collisions.

The gate has three bounded outcomes. `MECHANISM_ADMISSIBLE` would mean only that the registered rule meets its declared formal laws under the frozen frame. `FORMAL_MAP_REFUTED` would record a certified law violation for that rule. `SPLIT_IDEAL_CODOMAIN_OBSTRUCTION` would record failure specifically associated with literal split-branch compatibility. None is a theorem about every conceivable codomain, and no such outcome has been produced here. The strict frame remains fixed after review precisely so that an inconvenient outcome cannot be removed by redefining the target.

4.2 Gate Q: primitive-unoriented quotient completeness

Gate Q is a prospective population-construction contract. Every eligible inherited matrix row would need an exact loxodromic decision, a maximal group-element root witness, a level-(3) conjugacy class assignment, an inversion link, and one canonical unoriented owner ID. Positive relations require reproducible witnesses. Negative relations require a complete decision procedure or an equivalently auditable certificate; exhaustion of a bounded search is not enough. Deterministic serializers must make repeated runs byte-comparable.

The gate remains unresolved independently of Gate M. If exact completeness cannot be established from the object and algorithms, the outcome is `QUOTIENT_NOT_EVALUABLE`. If a predeclared resource limit is reached while relations remain unresolved, the outcome is `QUOTIENT_UNRESOLVED_STOP`. A complete quotient is not inferred from the number of rows, apparent matrix uniqueness, or a mechanism's labels. No quotient computation or certificate exists in this phase.

4.3 Prospective certificate graph and replay obligations

The architecture is most naturally read as a directed certificate graph. Its first node is an immutable object ledger. Each row would contain an exact projective matrix representation, a convention for lifts and central signs, evidence of level-(3) subgroup membership, and an exact loxodromic eligibility decision. A row that fails eligibility would leave the owner population under an explicit exclusion code rather than being silently discarded. The ledger would also bind the serializer version and canonical byte representation so that later witnesses refer to stable objects.

The second node is the quotient ledger. For an eligible row, a positive maximal-root claim would require both a root and a verified exponent relation in the frozen group. A claim of primitivity would require a complete negative decision against all admissible proper roots, not merely the absence of a root inside a bounded word ball. Likewise, a positive conjugacy or inversion link would carry a checkable conjugator, while a negative relation would require a decision method whose completeness and termination apply to the exact subgroup relation. Canonical unoriented IDs would be assigned only after transitivity, inversion closure, and representative independence had been audited. Population totals would reconcile input rows, exclusions, repetitions, oriented classes, and final unoriented owners.

The third node is the mechanism registry. A candidate would enter by immutable formula bytes together with its domain, codomain, source trace, normalization, and declared treatment of associates and split conjugate ideals. Tests of conjugacy invariance, inversion compatibility, and repetition behavior would be specified before any collision outcome was visible. Output validation would distinguish an actual nonzero prime ideal in $\mathbb{Z}[i]$ from an element, generator, norm, rational prime, factorization list, or Galois orbit. A failed test would attach to that exact registered candidate; it could not be generalized to unregistered mechanisms.

Only the fourth node would contain a performance ledger. Its population hash would have to match the certified quotient, and its mechanism hash would have to match a Gate-M-admissible registry entry. Baseline collisions, holdout membership, controls, and the definition of $S_H(M)$ would be frozen before outputs were exposed. A score produced from raw rows, an incomplete quotient, or a modified formula would be rejected as differently typed data rather than reported with a cautionary footnote. Thus provenance is part of the estimand, not merely administrative metadata.

Independent replay would verify each edge of this graph. It would check byte hashes, deserialize every object, replay positive witnesses, confirm the authority of negative decisions, reconcile population counts, and verify that performance code received only the authorized artifacts. Replay agreement would certify reproducibility of the registered computation, not truth beyond its formal and finite scope. Since none of these nodes or receipts exists in the present stage, the graph specifies an evidence target and its stop rules; it does not report a hidden prototype.

4.4 The limited meaning of independence

The two gates are independent in an architectural entailment sense only. Closure of Gate M does not entail closure of Gate Q because producing lawful labels does not decide primitive roots, conjugacy, or inversion classes. Closure of Gate Q does not entail closure of Gate M because a certified owner population does not define a literal ideal-selection law. Failure of either gate blocks downstream performance, but their error rates, computational costs, and empirical behavior have not been observed. The present article therefore makes no claim of mathematical, statistical, or empirical independence.

4.5 Performance firewall

Only after both gates pass could a prospectively frozen owner partition expose a finite collision-separation estimand such as `S_H(M)`. The statistic would count only predeclared baseline-collision owner pairs in a certified holdout for which a fixed mechanism returns different valid literal ideals. It could not choose the mechanism, repair a failed law, or compensate for duplicate/repeated owners. No value of `S_H` is reported. Even a positive future value would be a bounded finite-refinement result rather than a global orbit-to-prime-ideal theorem.

Typed controls would remain separate from the primary score. Permuting ideal labels without changing equivalence classes is a relabeling invariance test, not arithmetic specificity. Replacing literal primes with composite objects is a type violation, not a matched control. Directionally sensitive controls would need to be registered before output access and assessed under the same object, owner, and clock definitions.

5 Evidence-Synthesis Findings

The first registered finding is a support boundary. The frozen corpus identifies the Picard/Bianchi geodesic object and supplies primitive, inversion, group-algorithm, and ideal-arithmetic vocabulary. It does not define the project-specific Gaussian-prime-ideal owner law. Object identity and terminology therefore cannot be promoted into an owner map, a quotient, or a refinement score.

The second finding is that quotient completeness remains a composite certificate target. The adjacent methods do not arrive as one theorem or implemented solver for the exact level-(3) relation. The required hypothesis bridges, termination logic, positive and negative witnesses, and deterministic canonicalization remain prospective. This gap is genuine within the frozen design even if the literal codomain were later replaced in a different study.

The third finding is the review-adjudicated codomain interpretation. One literal Gaussian prime ideal is a deliberately strict project frame, not an intrinsic consequence of the geometry or literature. A split-branch obstruction under that frame would be conditional and construct-specific. An unordered Galois orbit or another equivariant output might avoid that problem, but considering it would require a separately frozen research question; it cannot be introduced as a repair after performance access.

The fourth finding is the pre-performance separation of mechanism admissibility and quotient completeness. Their design-level non-entailment prevents performance from selecting or repairing either interface. The contribution is the resulting fail-closed certificate architecture, not a theorem, novelty determination, implemented mechanism, or scientific result.

6 Reproducibility and Prospective Implementation Interface

Manuscript-level reproducibility is improved by naming the exact frozen artifacts that bound the argument: the Phase-2 source inventory and verification ledgers, the Phase-3 source-effect matrix and synthesis, the Phase-4 report and claim manifest, all four Phase-5 role reviews, the Phase-5 synthesis and checkpoint, and the Phase-6 claim-intent manifest. These artifacts preserve the screening counts, row-level source status, correction relation, claim-use boundaries, and review findings. They do not substitute for missing query strings or passage locators, and no claim of independent search reproducibility is made.

A later scientific implementation would require five auditable interfaces. First, an object serializer would encode projective matrices, subgroup membership evidence, and loxodromic eligibility. Second, a quotient engine would emit maximal-root, conjugacy, inversion, and canonicalization witnesses, including complete negative decisions. Third, a mechanism registry would freeze formula bytes, source trace, codomain, and law tests before any performance data are visible. Fourth, a control registry would preserve typed matched controls without changing the owner or clock. Fifth, an independent replay layer would verify hashes, population accounting, and deterministic receipts.

All five interfaces are prospective. No serializer, engine, registry entry, control result, or replay receipt is claimed to exist at Stage 2. Their purpose is to define what evidence would be needed for later evaluation and what stop state must be returned when evidence is missing.

7 Discussion and Implications

The strict-frame clarification changes the interpretation of the first gate without weakening the freeze. The literal codomain can be scientifically useful as a stress test of whether an unoriented geometric owner carries enough intrinsic data to select a non-Galois-stable branch. Yet failure would primarily speak to that deliberately demanding type choice. It must not be advertised as an obstruction to every Galois-stable or orientation-enriched ownership concept.

The quotient gate has a different status. It concerns the validity of the observational unit. Rows cannot become primitive owners through labeling, and finite performance on uncertified rows would not estimate the registered construct. This obligation persists under alternative codomains and is therefore independent of the split-frame question in the limited architectural sense defined above.

The combined architecture encourages interpretable negative outcomes. It distinguishes absence of a frozen admissible mechanism from formal refutation of one mechanism, and it distinguishes an unavailable exact quotient from a resource-bounded unresolved stop. Those distinctions prevent an engineering failure from being overread as a theorem and prevent a finite numerical success from concealing a type error.

The work may therefore be useful as a certificate-design protocol for arithmetic dynamics even while novelty and scientific performance remain unassessed. That value is methodological: it organizes what must be frozen, serialized, falsified, and replayed before a downstream score can be interpreted.

8 Acknowledged Limitations

The source study is bounded rather than exhaustive. It cannot support a literature-absence or novelty claim, and it may have missed a relevant owner construction. The evidence package does not contain full search strings, interface-level run logs, every screening decision, or a standalone reproducibility supplement. Existing upstream inventories and matrices improve traceability but do not close that editorial limitation.

All 22 paper-specific citation pairs, and all 144 citation pairs in the five-paper batch, lack passage locators. Consequently, statements near theorem boundaries have been narrowed to what the frozen source-verification summaries permit, but claim-to-passage support remains `INCONCLUSIVE`. No theorem-hypothesis transfer, maximal-root formula, conjugacy procedure, or split-ideal fact is certified at page or theorem level by this article. P29-S06/P29-S07 correction handling remains mandatory, structured retraction screening was not run, and source-level conflicts were not audited.

The contribution has not been positioned against the closest research-design literature through a separately authorized novelty analysis. The source retains nonprinting citation-provenance comments and visible status tokens because the Stage-2 audit contract requires them. Those elements support machine audit while leaving final passage adjudication to Stage 2.5.

No mechanism, quotient, owner population, collision set, `S_H` value, control result, or replay receipt exists here. No nonexistence theorem follows from the missing mechanism, and no universal split-prime obstruction is asserted. The literal codomain remains fixed for this study even though it is not claimed to be intrinsically privileged. The exact quotient remains independently unresolved.

The next authorization should also preserve a strict order of access. Passage adjudication should occur before a theorem is mapped to the object; the theorem-to-object map should be frozen before an implementation is evaluated; and both Gate M and Gate Q artifacts should be sealed before collision labels or holdout performance are exposed. This order is not a claim that one implementation strategy is uniquely correct. It is a safeguard against choosing definitions, stopping rules, or split branches because they improve a visible finite score.

A later manuscript could report partial progress without weakening this order. For example, it could close source locators while leaving both scientific gates open, instantiate a complete quotient while reporting no admissible mechanism, or refute one registered mechanism while leaving the broader search unresolved. Each outcome would require its own hashes and receipts and would retain the strict distinction between failure of a candidate and nonexistence in the whole hypothesis class. Such modular reporting is preferable to a single binary success label because it keeps the mathematical scope of every negative conclusion visible.

9 Future Work

The smallest source-level step would be a separately authorized locator pass for the few premises closest to an owner formula, group-root decision, and literal split-branch law. Any exact passage would need its hypotheses and prohibited transfers frozen; identity verification alone is insufficient.

The smallest object-level step would be a theorem-to-object map for projective lifts, subgroup membership, loxodromic status, maximal roots, level-(3) conjugacy, inversion, and deterministic serialization. It should specify stop states before implementation. A mechanism search and a quotient implementation should remain separate until both are independently frozen.

Only after those steps should a scholar-confirmed scientific design register a mechanism, owner partition, holdout, controls, resource limits, and replay protocol. Alternative Galois-stable or oriented codomains may be studied later as distinct hypotheses, not as post hoc replacements for the frozen literal target.

10 Conclusion

The frozen literature is sufficient to formulate a strict, auditable ownership study but not to produce an owner law or a certified owner population. The present article makes the construct boundary explicit: the literal Gaussian-prime-ideal codomain is deliberately strict and frozen, not canonical or forced, so any split obstruction is conditional on that frame. Quotient completeness is a separate unresolved problem. The two gates are independent only as non-entailing design obligations, and performance is downstream of both.

The article's defensible result is an evidence-synthesis and certificate architecture with explicit failure states and prospective interfaces. It is not a theorem, scientific computation, novelty finding, or Route advance. The formal Route-A tuple remains `UNASSIGNED`, positive arithmetic A2 remains `0/1`, and Route B remains uninvoked.

The immediate value of the architecture is diagnostic: it prevents a missing owner law, an incomplete observational quotient, and weak finite separation from being collapsed into one ambiguous outcome. A later study can close, refute, or stop at each interface with evidence appropriate to that interface. Until such artifacts exist, Gate M and Gate Q remain open and $S_H(M)$ remains undefined rather than zero.

Author Contributions

Liang Wang is the responsible human author. CRediT-style roles: Conceptualization; Methodology; Writing—review and editing; Project administration. He supplied the conceptual direction, made the author-adjudication decisions, approved the recorded workflow gates, interpreted the bounded evidence synthesis, and accepts final accountability for the manuscript. AI assistance is disclosed separately and is not credited with authorship.

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Competing Interests

The author declares no competing interests.

Ethics Statement

Ethics approval, informed consent, and human-subject review are not applicable. This article concerns mathematical objects, published-source records, and project-owned workflow artifacts; it reports no human participants, intervention, identifiable personal data, or human-related dataset.

Data and Materials Availability

The evidentiary materials are the frozen project artifacts identified in the manuscript. No new dataset, owner mechanism, quotient ledger, certificate, experiment, or numerical result was created for this Stage-2 composition.

AI-Assistance Disclosure and Verification Limitation

OpenAI Codex, operating in the GPT-5 model family, assisted during the session dated 2 September 2026 UTC; the exact backend snapshot or build was not exposed. Assistance covered organization of the frozen bibliography and source identities, bounded evidence synthesis, drafting, review integration, LaTeX conversion, and deterministic checks of identifiers, citations, references, hashes, and word counts. No AI-authored scientific data, experiment, mechanism, quotient, proof, or numerical result is presented. AI assistance is not credited with authorship.

Liang Wang made the recorded stage-gate confirmations and author-adjudicated framing decisions. Neither that responsibility nor the present composition implies human full-text or source-passage verification. Verification remains limited to frozen metadata, abstracts or authorized scope notes, source-verification ledgers, the source-effect matrix, and review artifacts. Exact passages were not frozen; all citation anchors therefore remain `none`, and every claim-to-passage assessment remains `INCONCLUSIVE` pending Stage 2.5.

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